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Power transmission device

Abstract

A power transmission device for a vehicle includes a housing that accommodates a power transmission mechanism, a control valve configured to control an oil pressure supplied to the power transmission mechanism, a first pump and a second pump configured to supply oil to the control valve. The housing includes a first chamber that accommodates the power transmission mechanism and the first pump, and a second chamber that is arranged adjacent to the first chamber in a horizontal line direction. In the first chamber, the first pump is arranged such that an axis of rotation of the first pump extends along a direction of an axis of rotation of the power transmission mechanism. In the second chamber, the control valve is arranged upright. In the second chamber, the second pump is arranged upright with an axis of rotation of the second pump extending along a vertical direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a U.S. national stage application of International Application No. PCT/JP2023/011608, filed on Mar. 23, 2023. This application also claims priority to Japanese Patent Application No 2022-047604, filed on Mar. 23, 2022.

BACKGROUND

Technical Field

(2) The present invention relates to a power transmission device.

Background Information

(3) Japanese Patent Laid-Open Publication No. 2015-45401 discloses a drive device for a vehicle.

SUMMARY

(4) In the case (housing) of this drive device (power transmission device) for a vehicle, a hydraulic control device (control valve) is arranged in an upright position. Further, an electric oil pump and a mechanical oil pump are disposed inside the case coaxially, aligned in the direction of the axis of rotation of the power transmission device.

(5) Therefore, the electric oil pump and the mechanical oil pump are disposed in the direction of the axis of rotation. When the electric oil pump and the mechanical oil pump are aligned in the direction of the axis of rotation, the part of the case that houses the electric oil pump and the mechanical oil pump becomes larger in the direction of the axis of rotation.

(6) Thus, there is a need to arrange the two pumps in the case without making the power transmission device larger in the direction of the axis of rotation.

(7) One aspect of the present disclosure is a power transmission device for a vehicle, comprising a housing that accommodates a power transmission mechanism, a control valve that controls the oil pressure supplied to the power transmission mechanism, and a first pump and a second pump that supply oil to the control valve, wherein the housing has a first chamber that accommodates the power transmission mechanism and the first pump, and a second chamber that is arranged adjacent to the first chamber in the horizontal line direction, in the first chamber, the first pump is arranged such that the axis of rotation of the first pump aligns in the direction of the axis of rotation of the power transmission mechanism, in the second chamber, the control valve is arranged upright, and in the second chamber, the second pump is arranged upright with the axis of rotation aligned in the vertical direction.

(8) According to one aspect of the present disclosure, since the axis of rotation of the first pump and the axis of rotation of the second pump are not coaxially arranged within the housing, the two pumps can be arranged within the housing without enlarging the power transmission device in the direction of the axis of rotation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Referring now to the attached drawings which form a part of this original disclosure.

(2) FIG. 1 is a schematic diagram illustrating the arrangement of the power transmission device in a vehicle.

(3) FIG. 2 is a schematic diagram illustrating a general configuration of the power transmission device.

(4) FIG. 3 is a schematic diagram illustrating the case from a second cover side.

(5) FIG. 4 is a schematic diagram illustrating the case from a first cover side.

(6) FIG. 5 is a schematic diagram illustrating the case from the front of the vehicle.

(7) FIG. 6 is a diagram illustrating a general configuration of an oil pressure control circuit.

(8) FIG. 7 is a schematic diagram illustrating a general configuration of the power transmission device according to a variant.

DETAILED DESCRIPTION

(9) First, definitions of terminology in the present specification will be explained.

(10) A power transmission device is a device having at least a power transmission mechanism, where the power transmission mechanism is, for example, at least one of a gear mechanism, a differential gear mechanism, or a reduction gear mechanism.

(11) In the following embodiment, a case is illustrated in which a power transmission device **1** has a function for transmitting the output rotation of an engine, but the power transmission device **1** need only transmit the output rotation of at least one of an engine or a motor (rotating electrical machine).

(12) “Overlaps as viewed from a prescribed direction” means that a plurality of elements are arranged in a prescribed direction, and means the same as “overlapping in a prescribed direction.” “Prescribed direction” is, for example, the axial direction, radial direction, direction of gravity, direction of travel of the vehicle (vehicle forward travel direction, vehicle rearward travel direction), etc.

(13) If a plurality of elements (parts, sections, etc.) are shown arranged in a prescribed direction in the drawings, it may be assumed that there is text in the description in the specification indicating that the plurality of elements overlap when viewed in the prescribed direction.

(14) “Not overlapping when viewed in a prescribed direction” and “offset when viewed in a prescribed direction” mean that a plurality of elements are not arranged in the prescribed direction, and mean the same as “not overlapping in a prescribed direction” and “offset in a prescribed direction.” “Prescribed direction” is, for example, the axial direction, radial direction, direction of gravity, direction of travel of the vehicle (vehicle forward travel direction, vehicle rearward travel direction), etc.

(15) If a plurality of elements (parts, sections, etc.) are shown not arranged in a prescribed direction in the drawings, it may be assumed that there is text in the description in the specification indicating that the plurality of elements do not overlap when viewed in the prescribed direction.

(16) “As viewed from a prescribed direction, a first element (part, section, etc.) is located between a second element (part, section, etc.) and a third element (part, section, etc.)” means that when viewed from the prescribed direction, the first element can be seen between the second element and the third element. The “prescribed direction” is, for example, the axial direction, radial direction, direction of gravity, direction of travel of the vehicle (forward movement direction of the vehicle, rearward movement direction of the vehicle), etc.

(17) For example, if the second element, the first element, and the third element are arranged in that order in the axial direction, then the first element is located between the second element and the third element when viewed from the radial direction. If the first element is shown between the second element and the third element as viewed from a prescribed direction in the drawings, it may be assumed that there is text in the description in the specification describing the first element between the second element and the third element as viewed from the prescribed direction.

(18) When two elements (parts, sections, etc.) overlap as viewed from the axial direction, the two elements are coaxial.

(19) “Axial direction” means the axial direction of the axis of rotation of a part making up the device. “Radial direction” means a direction orthogonally intersecting the axis of rotation of the part making up the device. The part is, for example, a motor, a gear mechanism, a differential gear mechanism, or the like.

(20) “The downstream side in the direction of rotation” means the downstream side in the direction of rotation when the vehicle is advancing or the direction of rotation during vehicle forward movement or during vehicle backward movement. It is appropriate to assume the downstream side in the direction of rotation is during vehicle forward movement, which is frequently the case.

(21) “Upright” with reference to the control valve means that in the case of a control valve having a basic configuration with a separation plate sandwiched between valve bodies, the valve bodies of

the control valve are stacked in the horizontal line direction based on the state of installation of the power transmission device in the vehicle. The “horizontal line direction” here does not mean the horizontal line direction in the strict sense, but also includes cases in which the direction of stacking is at an angle relative to the horizontal line.

(22) Further, “upright” with respect to the control valve means that the control valve is arranged with the plurality of pressure regulating valves inside the control valve aligned in the direction of a vertical line VL based on the state of installation of the power transmission device in the vehicle.

(23) “The plurality of pressure regulating valves aligned in the direction of a vertical line VL” means that the regulating valves inside the control valve are arranged spaced out in the direction of the vertical line VL.

(24) In this case, the plurality of pressure regulating valves need not be strictly arranged in single file in the direction of the vertical line VL.

(25) For example, if the plurality of valve bodies are stacked to form the control valve, the following is possible. Namely, the plurality of pressure regulating valves may be arranged in the direction of the vertical line VL with shifted positions in the direction of stacking of the valve bodies in the upright control valve.

(26) Further, as viewed from the axial direction of the valve bodies provided on the pressure regulating valves (the direction of forward and backward movement), the plurality of pressure regulating valves need not be arranged with gaps in between in the direction of the vertical line VL.

(27) As viewed from the axial direction of the valve bodies provided with the pressure regulating valves (the direction of forward and backward movement), the plurality of pressure regulating valves need not be adjacent in the direction of the vertical line VL.

(28) Hence, if, for example, the pressure regulating valves arranged in the direction of the vertical line VL are arranged with shifted positions in the stacking direction (horizontal line direction) of the valve bodies, then cases are also included in which, as viewed from the stacking direction, the pressure regulating valves that are adjacent in the direction of the vertical line VL are provided in a partially overlapping positional relationship.

(29) Further, that the control valve is “upright” means that the plurality of pressure regulating valves inside the control valve are arranged in the direction of movement of the valve bodies (spool valves) provided in the pressure regulating valves aligned in the horizontal line direction.

(30) The direction of movement of the valve bodies (spool valves) in this case is not limited to the horizontal line direction in the strict sense. The direction of movement of the valve bodies (spool valves) in this case is a direction along an axis of rotation X of the power transmission device. In this case, the direction of the axis of rotation X and the sliding direction of the valve bodies (spool valves) are the same.

(31) An embodiment of the present disclosure is described below.

(32) FIG. 1 is a schematic diagram illustrating the arrangement of a power transmission device **1** in a vehicle V.

(33) FIG. 2 is a schematic diagram illustrating a general configuration of the power transmission device **1**.

(34) As shown in FIG. 1, the power transmission device **1** in the front of the vehicle V is arranged between left and right frames FR, FR. A housing HS of the power transmission device **1** is configured by a case **6**, a first cover **7**, a second cover **8**, and a third cover **9**.

(35) As shown in FIG. 2, the interior of the housing HS accommodates a torque converter T/C, a forward-reverse switching mechanism **2**, a variator **3**, a reduction mechanism **4**, a differential drive device **5**, an electronic oil pump EOP, a mechanical oil pump MOP, a control valve CV, etc.

(36) In the power transmission device **1**, output rotation of an engine ENG (the drive source) is input into the forward-reverse switching mechanism **2** via the torque converter T/C.

(37) The rotation input to the forward-reverse switching mechanism **2** is forward rotation or reverse rotation and is input to a primary pulley **31** of the variator **3**.

(38) In the variator **3**, changing the winding radius of a belt **30** in the primary pulley **31** and a secondary pulley **32** causes the rotation input to the primary pulley **31** to be shifted at a desired gear ratio and output by an output shaft **33** of the secondary pulley **32**.

(39) Output rotation of the secondary pulley **32** is input via the reduction mechanism **4** into the differential drive mechanism **5** (differential gear mechanism) and is then transmitted to drive wheels WH, WH via left and right drive shafts **55A**, **55B**.

(40) The reduction mechanism **4** has an output gear **41**, an idler gear **42**, a reduction gear **43**, and a final gear **45**.

(41) The output gear **41** rotates together with the output shaft **33** of the secondary pulley **32**.

(42) The idler gear **42** meshes with the output gear **41** in a manner allowing transmission of rotation. The idler gear **42** is spline-fitted to an idler shaft **44** and rotates together with the idler shaft **44**. The idler shaft **44** is provided with the reduction gear **43** that has a smaller radius than the idler gear **42**. The reduction gear **43** meshes with the final gear **45** that is fixed to the outer circumference of a differential case **50** of the differential device **5** in a manner allowing transmission of rotation.

(43) In the power transmission device **1**, the forward-reverse switching mechanism **2**, the torque converter T/C, and the output shaft of the engine ENG are arranged coaxially (concentrically) along an axis of rotation X1 (first axis) of the primary pulley **31**.

(44) The output shaft **33** of the secondary pulley **32** and the output gear **41** are arranged coaxially along an axis of rotation X2 (second axis) of the secondary pulley **32**.

(45) The idler gear **42** and the reduction gear **43** are arranged coaxially along a common axis of rotation X3.

(46) The final gear **45** and the drive shafts **55A**, **55B** are arranged coaxially on a common axis of rotation X4. In the power transmission device **1**, the axes of rotation X1-X4 are set to have a positional relationship parallel to each other. In the following, these axes of rotation X1-X4 may, as needed, be referred to as the axis of rotation X of the power transmission device **1** (power transmission mechanism).

(47) FIG. **3** is a plan view of a case **6** as viewed from the second cover **8** side. FIG. **4** is a plan view of the case **6** as viewed from the first cover **7** side. Note that in the enlarged view of FIG. **3**, the strainer **10** and the mechanical oil pump MOP are omitted, showing the areas around the connecting sections **625**, **627** provided in a partition section **62**. In FIGS. **3** and **4**, hatching is used to clearly indicate the positions of the joining sections **611**, **612**.

(48) As shown in FIG. **3**, the case **6** has a cylindrical circumferential wall section **61** and the partition section **62**.

(49) The circumferential wall section **61** is arranged in the direction of the axis of rotation of a power transmission device **1** and forms the outer wall section of the power transmission device.

(50) The partition section **62** is provided within the circumferential wall section **61**, cutting across the axes of rotation (axes of rotation X1 to X4) of the power transmission mechanism.

(51) As shown in FIG. **2**, the partition section **62** divides the space inside the circumferential wall section **61** in two in the direction of the axis of rotation X1. One side of the partition section **62** in the direction of the axis of rotation X1 is a first chamber S1, and the other side is a third chamber S3.

(52) The first chamber S1 accommodates the forward-reverse switching mechanism **2**, the reduction mechanism **4**, and the differential device **5**. The third chamber S3 houses a variator **3**.

(53) In the case **6**, an opening on the first chamber S1 side is sealed by the second cover **8** (torque converter cover). An opening on the third chamber S3 side is sealed by the first cover **7** (side cover).

(54) In the case **6**, the oil used for the operation of the power transmission device **1** or lubrication of component elements of the power transmission device **1** is collected below the space between the first cover **7** and the second cover **8** (the first chamber S1 and the third chamber S3).

(55) As shown in FIG. 3, the end surface of the case **6** facing the second cover **8** side (toward the viewer) is a joining section **611** with the second cover **8**. The joining section **611** is a flanged section that surrounds the entire circumference of the opening on the second cover **8** side of the circumferential wall section **61**. A joining section **811** (see FIG. 2) on the second cover **8** side is joined to the entire circumference of the joining section **611**. The case **6** and the second cover **8** are connected by bolts, not shown, to join the joining sections **611**, **811** together. The opening of the case **6** is thus held in a state sealed by the second cover **8**, forming the closed first chamber **S1**.

(56) As shown in FIG. 3, the partition section **62** of the case **6** is provided in a direction essentially orthogonal to the axes of rotation (axes of rotation **X1** to **X4**). Through-holes **621**, **622**, **624** and a support hole **623** are formed in the partition section **62**.

(57) The through-hole **621** is formed around the axis of rotation **X1**. A cylindrical support wall section **631** surrounding the through-hole **621** and a circumferential wall section **641** surrounding the outer circumference of the cylindrical support wall section **631** with a space in between are provided in the surface of the partition section **62** facing the first chamber **S1** (toward the viewer). The cylindrical support wall section **631** and the circumferential wall section **641** project toward the viewer in FIG. 3 (toward the second cover **8** in FIG. 2).

(58) A region **651** between the support wall section **631** and the circumferential wall section **641** is a cylindrical space accommodating a piston (not shown) of the forward-reverse switching mechanism **2**, friction plates (forward clutch, reverse brake), etc.

(59) An input shaft **34** (see FIG. 2) of the primary pulley **31** is rotatably supported on the inner circumference of the cylindrical support wall section **631** via bearings **B**.

(60) As shown in FIG. 3, the through-hole **622** is formed around the axis of rotation **X2**.

(61) In the power transmission device **1** mounted in the vehicle **V**, the axis of rotation **X2** is positioned diagonally above toward the rear of the vehicle as viewed from the axis of rotation **X1**.

(62) A cylindrical support wall section **632** that surrounds the through-hole **622** is provided in the surface of the partition section **62** facing the first chamber **S1** (toward the viewer). The support wall section **632** projects toward viewer in FIG. 3 (toward the second cover **8** in FIG. 2).

(63) The **631** rotatably supports the output shaft **33** of the secondary pulley **32** (see FIG. 2) via the bearings **B**.

(64) As shown in FIG. 3, the support hole **623** is a closed-bottom hole formed around the axis of rotation **X3**.

(65) In the power transmission device **1** mounted in the vehicle **V**, the axis of rotation **X3** is positioned diagonally above toward the rear of the vehicle when viewed from the axis of rotation **X1**, and diagonally below toward the rear of the vehicle when viewed from the axis of rotation **X2**.

(66) On the surface of the partition section **62** on the first chamber **S1** side (toward the viewer), a cylindrical support wall section **633** that surrounds the support hole **623** is provided. In FIG. 3, the support wall section **633** protrudes toward the viewer (the second cover **8** side in FIG. 2). The support wall section **633** surrounds the outer circumference of the support hole **623** with a gap therebetween. The inner circumference of the support wall section **633** supports one end of the idler shaft **44** of the reduction mechanism **4** (see FIG. 2) rotatably via the bearings **B**.

(67) As shown in FIG. 3, the through-hole **624** is formed centered on the axis of rotation **X4**.

(68) In the power transmission device **1** mounted in the vehicle **V**, the axis of rotation **X4** is positioned diagonally below toward the rear of the vehicle when viewed from the axis of rotation **X1**, diagonally below toward the rear of the vehicle when viewed from the axis of rotation **X2**, and diagonally below toward the front of the vehicle when viewed from the axis of rotation **X3**.

(69) On the surface of the partition section **62** on the first chamber **S1** side (toward the viewer), a cylindrical support wall section **634** that surrounds the through-hole **624** is provided. In FIG. 3, the support wall section **634** protrudes toward the viewer (the second cover **8** side in FIG. 2). The support wall section **634** surrounds the outer circumference of the through-hole **624** with a gap therebetween. The inner circumference of the support wall section **634** rotatably supports the

differential case **50** of the differential device **5** (see FIG. 2) via the bearings **B**.

(70) As shown in FIG. 2, the final gear **45**, in the form of a ring as viewed from the direction of the axis of rotation **X4**, is fixed to the outer circumference of the differential case **50**. The final gear **45** rotates about the axis of rotation **X4** together with the differential case **50**.

(71) In the case **6** shown in FIG. 3, in a region below the arcuate circumferential wall section **641** and farther toward the front of the vehicle than the final gear **45**, a housing section **67** is provided where the strainer **10** and the mechanical oil pump MOP (first pump) are arranged.

(72) The strainer **10** is arranged in a position intersecting a straight line **Lm** connecting the outer circumference of the circumferential wall section **641** and the outer circumference of the final gear **45**. The strainer **10** is arranged upwards toward the side of the circumferential wall section **641** while avoiding interference with the final gear **45**.

(73) The mechanical oil pump MOP is arranged toward the front side of the vehicle (right side in the diagram) as viewed from the strainer **10**. The mechanical oil pump MOP is provided such that the axis of rotation **X5** of the mechanical oil pump MOP is aligned in the direction of the axes of rotation **X1** to **X4** of the power transmission mechanism.

(74) A connection section **625** with the strainer **10** and a connection section **627** with the mechanical oil pump MOP are provided in the partition section **62** are provided below the circumferential wall section **641**, **62**, as shown in FIG. 3.

(75) A connection port **625a** of the connection section **625** and a connection port **627a** of the connecting section **627** open in the same direction. The connection port **625a** of the connection section **625** communicates with an oil path **626** provided in the partition section **62**. The connection port **627a** of the connecting section **627** communicates with an oil path **628** provided in the partition section **62**.

(76) The oil paths **626**, **628** extend in a straight line toward a housing section **68** (to the right in the diagram) inside the partition section **62**. The oil path **626** connects to the electric oil pump EOP housed in the housing section **68** (housing chamber **S2**) via the oil paths inside case **6**. The oil path **628** connects to the control valve **CV** (see FIG. 2) installed in the housing section **68** via the oil paths inside the case **6**.

(77) As shown in FIG. 4, in the surface of case **6** on the first cover **7** side, the region corresponding to the oil path **626** bulges toward the viewer (the first cover **7** side) below the circumferential wall section **61** that surrounds the third chamber **S3**.

(78) On the end surface of the circumferential wall section **61** on the first cover **7** side, a joining section **612** with the first cover **7** is provided. The joining section **612** is a flanged section surrounding the entire opening of the partition section **62** facing the first cover **7**. The inside of the joining section **612** is the partition section **62**.

(79) The joining section **711** (see FIG. 2) on the first cover **7** side is joined to the entire circumference of the joining section **612**. The case **6** and the first cover **7** are connected by bolts, not shown, to join the joining sections **612**, **712** together. The opening of the case **6** is thereby maintained in a sealed state by the first cover **7**, forming the closed third chamber **S3**.

(80) As shown in FIG. 4, the through-hole **622** is positioned diagonally above the rear side of the vehicle as viewed from the through-hole **621**.

(81) Inside the circumferential wall section **61**, the primary pulley **31** and the secondary pulley **32** of the variator **3** are located in the upper region in the direction of the vertical line **VL**.

(82) The circumferential wall section **61** is such that the bottom region where the primary pulley **31** is provided bulges significantly below the case **6**, and a bulging region **600** corresponding to the oil path **626** is located below this bulging region.

(83) As viewed from the direction of the axis of rotation **X1**, the bulging region **600** is provided in a range that extends across the circumferential wall section **61** on the side toward the front of the vehicle to the housing section **68**.

(84) As shown in FIG. 2, the housing section **68** is attached to a side surface of the case **6** toward

the front of the vehicle.

(85) The housing section **68** is provided with an opening facing the front side of the vehicle. The housing section **68** is provided along the axis of rotation **X1**. As viewed from the radial direction of the axis of rotation **X1**, the housing section **68** is formed having a range in the direction of the axis of rotation **X1** from the region of the circumferential wall section **61** of the case **6** to the side of the first cover **7**.

(86) A region of approximately half of a bottom wall section **682** of the housing section **68** on the engine **ENG** side forms a single unit with the circumferential wall section **61**. The region of approximately half of the opposite side of the bottom wall section **682** is an extension of the circumferential wall section **61**, with a gap formed with the outer circumference of the first cover **7**.

(87) FIG. 5 is a diagram of the housing section **68** as viewed from the front of the vehicle. Note that in FIG. 5, crosshatching is applied to a joining section **683** to clearly distinguish the location of the joining section **683** in a surrounding wall **681**.

(88) The housing section **68**, as viewed from the front of the vehicle, has the surrounding wall **681** encompassing the entire outer circumference of the bottom wall section **682**. The end surface of the surrounding wall **681** facing the viewer serves as the joining section **683** with the third cover **9**. The joining section **683** is a flanged section surrounding the entire circumference of the opening of the surrounding wall **681** on the third cover **9** side.

(89) As shown in FIG. 2, a joining section **911** on the third cover **9** side is joined to the entire circumference of the joining section **683**. The housing section **68** and the third cover **9** are connected by bolts, not shown, with the joining sections **683**, **911** joined to each other. The opening of the housing section **68** is thereby maintained in a sealed state by the third cover **9**, forming the closed second chamber **S2** (second chamber).

(90) As shown in FIG. 5, the housing chamber **S2** accommodates the control valve **CV** and the electric oil pump **EOP**.

(91) In the housing chamber **S2**, the control valve **CV** and the electric oil pump **EOP** are arranged aligned with the direction of the axis of rotation **X** of the power transmission device **1** (power transmission mechanism).

(92) As shown in FIG. 5, inside the housing chamber **S2**, the control valve **CV** is upright, so that the direction of stacking of the valve bodies **921**, **921** is aligned in the front-rear direction of the vehicle (toward and away from the viewer in the figure).

(93) In the housing chamber **S2**, the control valve **CV** is upright so as to satisfy the following conditions. (a) A plurality of pressure regulating valves **SP** (spool valves) inside the control valve **CV** are aligned in the direction of the vertical line **VL** (the vertical direction) based on the state of installation of the power transmission device **1** in the vehicle **V**, and (b) a direction of advancement and retraction **Xp** of the pressure regulating valves **SP** (spool valves) is along the horizontal line direction.

(94) Advancement and retraction of the pressure regulating valves **SP** (spool valves) is thus not hindered, and the control valve **CV** is upright in the housing chamber **S2**. Thus, the housing chamber **S2** is prevented from becoming larger in the front-rear direction of the vehicle.

(95) As shown in FIG. 5, the control valve **CV**, as viewed from the front of the vehicle, forms an approximate L-shape provided with a cut-out **923** in the essentially rectangular valve body **921**. The cut-out **923** in the housing chamber **S2** is located under the region overlapping the first cover **7**.

(96) As viewed from the front of the vehicle, the cut-out **923** is a region formed by removing an area straddling an edge **921a** and the lower edge **921b** on the electric oil pump **EOP** side of the control valve **CV**.

(97) In the control valve **CV**, providing the cut-out **923** forms a side edge **921a'** offset from edge **921a** to the opposite side (the edge **921c** side), and an offset side edge **921b'** from the lower edge **921b** to the upper side (the edge **921d** side).

(98) These edges **921a**, **921b'** and an imaginary line La along the edge **921a** and an imaginary line Lb along the edge **921b** are the region of the cut-out **923**. As viewed from the front of the vehicle, in the control valve CV, the cut-out **923** is formed as a region that can accommodate part of the electric oil pump EOP.

(99) At least part of the electric oil pump EOP is accommodated in the cut-out **923** as viewed from the front of the vehicle.

(100) The electric oil pump EOP has a basic configuration in which a control unit **931**, a motor unit **932**, and a pump unit **933** are arranged in a straight line in the direction of an axis of rotation Z1 of the motor.

(101) The electric oil pump EOP is provided such that the axis of rotation Z1 is orthogonal to the axis of rotation X of the power transmission device **1**. In this state, the axis of rotation Z1 of the electric oil pump EOP is arranged in the direction of the vertical line VL (the vertical direction). Further, in the electric oil pump EOP, the pump unit **933** is located at the very bottom of the housing chamber S2. An intake port **933a** and a discharge port **933b** of the pump unit **933** are located on the boundary toward the motor unit **932** and are each connected to an oil path inside the case.

(102) The intake port **933a** is connected to the strainer **10** via the oil path **626** (see FIG. 3) inside the partition section **62** described above.

(103) The strainer **10** is housed in the first chamber S1, separately from the housing chamber S2 of the control valve CV (see FIG. 3). The mechanical oil pump MOP, to which the strainer **10** is attached, is also housed in the first chamber S1.

(104) As shown in FIG. 5, as viewed from the front of the vehicle, the strainer **10** and the mechanical oil pump MOP are arranged at the position indicated by the dashed lines, away from the viewer behind the housing chamber S2. As viewed from the front of the vehicle, the strainer **10** and the mechanical oil pump MOP are provided in a positional relationship overlapping the control valve CV.

(105) In the present embodiment, by positioning the pump unit **933** of the electric oil pump EOP at the very bottom of chamber S2, the vertical line VL direction position of the intake port **933a** of the pump unit **933** is brought closer to the strainer **10**.

(106) The length of the oil path connecting the strainer **10** and the intake port **933a** of the electric oil pump EOP is thereby reduced.

(107) The upper side of the control valve CV extends up to above the electric oil pump EOP. As viewed from the direction of the vertical line VL (the direction of the axis of rotation Z1 of the electric oil pump EOP), the electric oil pump EOP is provided in a positional relationship overlapping the control valve CV.

(108) The frame FR of the vehicle V is located on the side of the first cover **7**, and there is no room to expand the housing HS in the vehicle width direction (the left-right direction in the diagram).

(109) In the present embodiment, when the electric oil pump EOP is disposed in the housing chamber S2, instead of simply aligning the electric oil pump parallel with the control valve CV, the electric oil pump EOP is arranged in the cut-out **923** provided in the control valve CV.

(110) Consequently, the electric oil pump EOP is arranged so that part of the frame FR protrudes toward the frame FR side from the cut-out **923**. A protrusion width Wx is equal to a width W3 of the electric oil pump EOP in the direction of the axis of rotation X minus a width W2 of the cut-out **923** in the direction of the axis of rotation X ($Wx=W3-W2$).

(111) Therefore, compared to simply aligning the control valve CV and the electric oil pump EOP in the direction of the axis of rotation X of the power transmission mechanism, the width of the housing chamber S2 can be reduced by the width W2 of the cut-out **923**.

(112) Thus, (a) by arranging the control valve CV upright, the housing chamber S2 can be prevented from increasing in size in front-rear direction of the vehicle. (b) By utilizing the cut-out **923** in the control valve CV, the housing chamber S2 can be prevented from increasing in size in

the width direction of the vehicle.

(113) By arranging both the control valve CV and the electric oil pump EOP upright within chamber S2, the housing chamber S2 can be prevented from increasing in size in the front-rear direction and the width direction of the vehicle.

(114) Consequently, when the control valve CV and the electric oil pump EOP are arranged in the housing chamber S2 (the housing section 68), which is different from the first chamber S1, within the case 6, the housing HS of the power transmission device 1 can be prevented from increasing in size in the front-rear direction and the width direction of the vehicle.

(115) The oil pressure control circuit 95 inside the control valve CV regulates the operating oil pressure of the power transmission mechanism (the forward-reverse switching mechanism 2, the variator 3, etc.) based on the oil pressure generated by the oil pump.

(116) The power transmission device 1 is equipped with one each of the mechanical oil pump MOP and the electric oil pump EOP as oil pumps. These oil pumps suction the oil OL collected in the bottom of the housing HS through the strainer 10, pressurize the suctioned oil OL, and supply the oil OL to an oil pressure control circuit 95 (see FIG. 6) inside the control valve CV.

(117) FIG. 6 is a diagram illustrating an example of the oil pressure control circuit 95 inside the control valve CV, showing parts related to regulation of the operating oil pressure of the variator 3 (the primary pulley 31, the secondary pulley 32) within the oil pressure control circuit 95.

(118) A first pressure regulating valve 951 adjusts the line pressure based on the oil pressure generated by the oil pump OP by regulating the drainage amount of the oil OL in the first pressure regulating valve 951.

(119) The line pressure adjusted by the first pressure regulating valve 951 is supplied to a primary pressure regulating valve 953, a secondary pressure regulating valve 954, and a second pressure regulating valve 952. Note that the line pressure is also supplied to other pressure regulating valves provided in the oil pressure control circuit 95.

(120) The second pressure regulating valve 952 adjusts the pilot pressure based on the line pressure.

(121) The pilot pressure adjusted by the second pressure regulating valve 952 is supplied to a primary solenoid 955 and a secondary solenoid 956.

(122) The primary solenoid 955 and the secondary solenoid 956 operate based on instructions from a control device (not shown), regulating the signal pressure supplied to the primary pressure regulating valve 953 and the secondary pressure regulating valve 954.

(123) In the primary pressure regulating valve 953 and the secondary pressure regulating valve 954, the spool valves move in the axial direction (the Xp direction in the diagram) in response to the signal pressure.

(124) When the spool valves move in the axial direction, the line pressure regulated by the first pressure regulating valve 951 is further regulated to a pressure corresponding to the position of the spool valves, and then supplied to pressure chambers of the corresponding pulleys (the primary pulley 31, the secondary pulley 32).

(125) The first pressure regulating valve 951 is located at the farthest upstream side (the oil pump OP side) in the oil pressure control circuit 95, where the oil pressure generated by the oil pump OP is first supplied.

(126) In the first pressure regulating valve 951, part of the oil produced by the oil pump OP is drained during adjustment of the line pressure based on the oil pressure generated by the oil pump OP.

(127) The first pressure regulating valve 951 is provided inside the control valve CV, and an oil discharge hole 951a from which the oil OL drained from the first pressure regulating valve 951 is discharged opens in the bottom of the control valve CV.

(128) The control valve CV needs to be arranged so that the direction Xp of advancement and retraction of the pressure regulating valves (spool valves) within the oil pressure control circuit 95

is aligned with the horizontal line HL direction (see FIG. 5).

(129) The control valve CV is therefore arranged upright within the housing chamber S2.

(130) As shown in FIG. 5, in the control valve CV, the maximum width W2 of the cut-out 923 in the horizontal line direction is set so that enough of the width W1 is left to dispose the pressure regulating valves (spool valves) in the horizontal line direction to the side of the cut-out 923.

(131) The edge 921a of the control valve CV on the electric oil pump EOP side is positioned beyond the first cover 7 relative to the axis of rotation Z1 of the electric oil pump EOP.

(132) Thus, within the space of the housing chamber S2, almost all regions except the region needed for the installation of the electric oil pump EOP are disposed to overlap the control valve CV, as viewed from the front of the vehicle. In other words, in addition to the horizontal arrangement of the pressure regulating valves (spool valves), the cut-out 923 is provided to ensure that sufficient volume of the control valve CV is maintained for the installation of the electric oil pump EOP.

(133) As described above, the power transmission device 1 for a vehicle according to the present embodiment has the following configuration.

(134) (1) The power transmission device 1 comprises a power transmission mechanism (the torque converter T/C, the forward-reverse switching mechanism 2, the variator 3, the reduction mechanism 4, the differential device 5) that transmits drive power from the engine ENG (drive source) to the drive wheels WH, WH, a housing HS that accommodates the power transmission mechanism, a control valve CV that controls the oil pressure supplied to the power transmission mechanism, a mechanical oil pump MOP (first pump) and an electric oil pump EOP (second pump) that supply oil OL to the control valve CV.

(135) The housing HS comprises a first chamber S1 that accommodates the power transmission mechanism and the mechanical oil pump MOP, and a second chamber S2 (housing chamber S2) arranged adjacent to the first chamber S1 in the horizontal line HL direction.

(136) In the first chamber S1, the electric oil pump EOP is arranged so that the axis of rotation X5 of the electric oil pump EOP is aligned with the axis of rotation X of the power transmission mechanism.

(137) In the housing chamber S2, the control valve CV is arranged upright.

(138) In the housing chamber S2, the electric oil pump EOP is arranged upright so that the axis of rotation Z1 of the electric oil pump EOP is aligned in the vertical direction (direction of the vertical line VL).

(139) With this configuration, since the axis of rotation X5 of the mechanical oil pump MOP and the axis of rotation Z1 of the electric oil pump EOP are not coaxially arranged within the housing HS, the two pumps can be arranged within the housing HS without enlarging the power transmission device 1 in the direction of the axis of rotation X.

(140) (i) The housing HS includes the case 6 inside which the first chamber S1 is formed, the housing section 68 attached to the side of case 6 toward the front of the vehicle when the power transmission device 1 is mounted in the vehicle V, and the third cover 9 that seals the opening of the housing section 68 from the front of the vehicle.

(141) The housing chamber S2 is formed between the circumferential wall section 61 of case 6, the housing section 68, and the third cover 9.

(142) The first chamber S1 and the housing chamber S2 (second chamber) are adjacent, separated by the circumferential wall section 61 of the case 6.

(143) With this configuration, the control valve CV and the electric oil pump EOP are accommodated in the housing chamber S2 (second chamber), a separate space from the first chamber S1.

(144) Thus, extra space is created in the case 6, thereby improving the layout within the housing HS (the case 6).

(145) If the control valve CV is provided at the bottom of the housing HS, the power transmission

device will be larger in the direction of the vertical line VL by an amount equal to the size of the control valve CV.

(146) By installing the control valve CV in the housing chamber S2 provided in the housing HS toward the front of the vehicle, the size of the power transmission device **1** in the direction of the vertical line VL can be limited.

(147) Further, when the control valve CV is installed in the housing chamber S2 provided in the housing HS toward the front of the vehicle, it is necessary to prevent the power transmission device **1** from becoming larger toward the front of the vehicle.

(148) Therefore, the control valve CV is arranged upright so that the direction of advancement and retraction of the pressure regulating valves within the oil pressure control circuit **95** is aligned in the horizontal line direction. Arranging the control valve CV upright can minimize the extent to which the power transmission device **1** becomes larger toward the front of the vehicle.

(149) (2) The control valve CV and the electric oil pump EOP are arranged in the housing chamber S2 in the direction of the axis of rotation X of the power transmission mechanism.

(150) If the control valve CV and the electric oil pump EOP were aligned in the radial direction of the axis of rotation X of the power transmission mechanism, the power transmission device **1** would become larger in the radial direction.

(151) As the control valve CV and the electric oil pump EOP are aligned in the direction of the axis of rotation X, the power transmission device **1** can be prevented from becoming larger in the radial direction.

(152) For example, if the housing chamber S2 is located in the power transmission mechanism toward the front of the vehicle along the axis of rotation X, the power transmission device **1** can be prevented from becoming larger in the radial direction. In this case, the direction of the axis of rotation X corresponds to the vehicle width direction. The body frame FR and the engine ENG (drive source) are positioned on both sides of the power transmission device **1** in the vehicle width direction without extra space. Since there is relatively more space on the side of the power transmission device **1** toward the front of the vehicle, arranging the control valve CV and the electric oil pump EOP upright in the front-side housing chamber S2 allows the power transmission device **1** to be installed in the vehicle V without significantly affecting the vehicle layout.

(153) (3) The housing chamber S2 (housing chamber S2) is located toward the front of the vehicle as viewed from the first chamber S1.

(154) As viewed from the front of the vehicle, the control valve CV has the cut-out **923**.

(155) As viewed from the front of the vehicle, at least part of the electric oil pump EOP is positioned within the cut-out **923** of the control valve CV.

(156) If at least part of the electric oil pump EOP is positioned within the cut-out **923** of the control valve CV, the width (range) of the housing chamber S2 in the direction of the axis of rotation X can be minimized.

(157) As viewed from the front of the vehicle, the width in the direction of the axis of rotation X can be minimized by an amount equal to the positioning the electric oil pump EOP within the cut-out **923**.

(158) If a cut-out were not provided in the control valve CV, the control valve CV and the electric oil pump EOP would simply be arranged in the direction of the axis of rotation X. Utilizing the cut-out **923** makes it possible to reduce the volume of the housing chamber S2 in the direction of the axis of rotation X (increasing in size in the vehicle width direction) needed to install the control valve CV and the electric oil pump EOP compared to if there were no cut-out.

(159) This allows for a more compact accommodation of the control valve CV and the electric oil pump EOP within the housing section **68**.

(160) (4) As viewed from the front of the vehicle, the cut-out **923** is provided below the control valve CV in the direction of the vertical line VL, based on the installation state of the power transmission device **1** in vehicle V.

(161) As viewed from one side in the vertical line VL direction (vertical direction), the electric oil pump EOP and the control valve CV are positioned in an overlapping relationship.

(162) With this configuration, as viewed from the direction of the vertical line VL, at least part of the electric oil pump EOP is positioned within the housing chamber S2 to overlap the control valve CV.

(163) This allows the control valve CV and the electric oil pump EOP to be accommodated within the housing section **68**.

(164) Additionally, as viewed from the front of the vehicle, the strainer **10** positioned in the bottom of the case **6** is arranged in the first chamber **S1**, which is located behind the housing chamber **S2**. As viewed from the front of the vehicle, the strainer **10** is arranged in a positional relationship overlapping the lower region of the housing chamber **S2**. Therefore, the electric oil pump EOP can be positioned closer to the oil reservoir at the bottom of the case **6**.

(165) The length of the oil path within the case connecting the strainer **10** and the electric oil pump EOP can thereby be made shorter. A longer oil path in the case increases the resistance (suction resistance) when the electric oil pump EOP draws the oil OL through the strainer **10**. By shortening the length of the oil path in the case, a reduction in suction resistance can be expected.

(166) (5) The electric oil pump EOP comprises a motor unit **932** that houses a motor, and a pump unit **933** that suctions and pressurizes oil OL using the rotational force of the motor.

(167) The motor unit **932** and the pump unit **933** are aligned in the direction of the axis of rotation **Z1** of the electric oil pump EOP.

(168) As viewed from the front of the vehicle, the electric oil pump EOP is positioned such that the axis of rotation **Z1** of the motor is orthogonal to the axis of rotation **X** of the power transmission mechanism.

(169) With this configuration, the electric oil pump EOP can be arranged without enlarging the housing chamber **S2** in the direction of the axis of rotation **X**.

(170) (6) The pump unit **933** is located below the motor unit **932**.

(171) With this configuration, the pump unit **933** can be arranged at the very bottom in the direction of the vertical line VL within the housing chamber **S2**. In this case, as viewed from the front of the vehicle, the oil intake port **933a** of the pump unit **933** can be brought closer to the strainer **10** disposed at the bottom of the housing **HS**.

(172) The length of the oil path within the case connecting the strainer **10** and the oil intake port **933a** of the oil OL can thus be made even shorter, which can lead to a further reduction in suction resistance.

(173) (7) The first chamber **S1** has a strainer **10** arranged in the bottom of the first chamber **S1**.

(174) The mechanical oil pump MOP (first pump) and the electric oil pump EOP (second pump) share the strainer **10**.

(175) The strainer **10** is arranged in the first chamber **S1**, separate from the housing chamber **S2** of the control valve CV. Disposing the control valve CV is not disposed in the first chamber **S1** increases the freedom of layout in the first chamber **S1**.

(176) Additionally, the strainer **10** can be arranged in an optimal location within the first chamber **S1**, without having to consider the positional relationship with the control valve CV, thereby enhancing the flexibility of arrangement of the strainer **10**.

(177) For example, since the strainer **10** is smaller than the electric oil pump EOP or the control valve CV, the strainer can be arranged utilizing the gaps within the case **6**.

(178) For instance, in the case **6**, there is extra space in the lower region between the rotating bodies, which are components of the forward-reverse switching mechanism **2**, and the final gear **45**. This space (the housing section **67**) can accommodate the strainer **10**, further limiting the size of the case **6** in the direction of the vertical line VL.

(179) (8) As viewed from the front of the vehicle, the control valve CV is arranged in an overlapping relationship with the mechanical oil pump MOP and the strainer **10**.

(180) With this configuration, the control valve CV, the mechanical oil pump MOP, and the strainer **10** are arranged aligned in the front-rear direction of the vehicle. A width W_max of the housing chamber S2 need only accommodate the width W1 of the control valve CV plus the width Wx by which the upright electric oil pump EOP extends beyond the cut-out **923** in the direction of the axis of rotation X.

(181) The electric oil pump EOP can thus be disposed in the housing chamber S2, minimizing the extent to which the housing chamber S2 increases in size in the direction of the axis of rotation X.

(182) (ii) As viewed from the front of the vehicle, the electric oil pump EOP is positioned in a location overlapping the first cover **7** in the housing chamber S2 (case section). The control valve CV is positioned in a location overlapping the case **6** in the housing chamber S2.

(183) The pressure regulating valves inside the control valve CV are arranged in multiple rows in the vertical line VL direction, aligned in the vehicle width direction.

(184) The frame FR of the vehicle V is located on the first cover **7** side of the power transmission device **1**.

(185) Therefore, it is not preferable to enlarge the housing section **68** in the direction of the first cover **7** (the right side in FIG. 5).

(186) Therefore, the electric oil pump EOP is arranged upright in a location overlapping the first cover **7** in the housing chamber S2, and the region of the control valve CV that houses the pressure regulating valves (spool valves) is arranged in a location overlapping the case **6**.

(187) The housing section **68** can thereby be appropriately prevented from increasing in size in the direction of the first cover **7** (right side in FIG. 5).

(188) To arrange the pressure regulating valves inside the control valve CV in a direction aligned with the horizontal line HL, arranging the control valve CV horizontally within the housing chamber S2 would make the power transmission device **1** larger in the vehicle front-rear direction.

(189) Therefore, in the foregoing embodiment, the control valve CV is arranged upright, and the pressure regulating valves (spool valves) are aligned in the direction of the horizontal line HL and arranged in multiple rows in the direction of the vertical line VL. The extent to which the power transmission device **1** increases in size in the vehicle front-rear direction can thereby be minimized.

(190) By arranging the electric oil pump EOP in the cut-out **923** of the control valve CV within the housing section **68**, the extent to which the power transmission device **1** increases in size in the direction of the axis of rotation X (vehicle width direction) is minimized. The area around the power transmission device **1** in the vehicle V has strict limitations in the vehicle width direction.

(191) Therefore, by arranging the control valve CV upright and positioning the electric oil pump EOP in the cut-out **923** of the control valve CV, the power transmission device **1** can be appropriately positioned within these strict limitations.

(192) In the foregoing embodiment, a case was illustrated in which the power transmission device **1** transmits rotation of the engine ENG to the drive wheels WH, WH, but it is also possible for the power transmission device **1** to transmit the rotation of at least one of the engine ENG or the motor (rotating electrical machine) to the drive wheels WH, WH. For example, it is possible to use a single-motor, double-clutch power transmission device (in which the motor is arranged between the engine ENG and the power transmission device, a first clutch is disposed between the engine ENG and the motor, and a second clutch is arranged inside the power transmission device **1**).

(193) Further, in the foregoing embodiment, a case was illustrated in which the power transmission device **1** has a shifting function, but it is also possible for the power transmission device simply to reduce speed (or increase speed) without a shifting function. If the power transmission device does not have a shifting function and instead reduces and transmits rotation of the motor to the drive wheels WH, WH, the oil pressure control circuit for supplying the oil OL for cooling the motor and the oil OL for lubricating the reduction mechanism is arranged in the second chamber S2 with the electric oil pump EOP. Further, in the foregoing embodiment, a case was illustrated in which the control unit of the power transmission device **1** was provided with the control valve CV, but if the

power transmission device **1** does not have a shifting mechanism and the drive source is a motor (rotating electrical machine) and not the engine ENG, then the control unit may be provided with an inverter or the like for controlling the driving of the motor.

(194) FIG. **7** is a schematic diagram illustrating the general configuration of a modified example of the power transmission device.

(195) In the foregoing embodiment, a case was illustrated in which the first chamber **S1** housing the strainer **10** and the housing chamber **S2** utilizing the electric oil pump EOP and the control valve CV are divided by the bottom wall section **682** of the housing section **68**, which is integral with the circumferential wall section **61**. As shown in FIG. **7**, a power transmission device **1A** using a case **6A**, in which the first chamber **S1** and the housing chamber **S2** communicate through an opening **684**, is also possible.

(196) In this power transmission device **1A**, the control valve CV is arranged to block the opening **684** and partition the first chamber **S1** and the housing chamber **S2**.

(197) Even with the power transmission device **1A** having this configuration, the power transmission device **1A** can be prevented from increasing in size, while creating extra space within the housing HS, thereby improving the layout characteristics inside the housing HS.

(198) Embodiments of the present invention have been described above, but the present invention is not limited to those aspects shown in the embodiments. The present invention may be appropriately modified within the scope of the technical concept of the invention.

DESCRIPTION OF THE REFERENCE SYMBOLS

(199) **1, 1A** Power transmission device (power transmission mechanism); **2** Forward-reverse switching mechanism (power transmission mechanism); **3** Variator (power transmission mechanism); **4** Reduction mechanism (power transmission mechanism); **5** Differential device (power transmission mechanism); **6** Case; **668** Housing chamber (case section); **10** Strainer; **923** Cut-out; **932** Motor unit; **933** Pump unit; CV Control valve; MOP Mechanical oil pump (first pump); EOP Electric oil pump (second pump); HS Housing; **S1** First chamber; **S2** Second chamber; T/C Torque converter (power transmission mechanism); ENG Engine (drive source); X, **X1-X4**, **Z1** Axes of rotation; and **X5** Axis of rotation (axis of the first pump).

Claims

1. A power transmission device for a vehicle, comprising: a housing that accommodates a power transmission mechanism; a control valve configured to control an oil pressure supplied to the power transmission mechanism; and a first pump and a second pump configured to supply oil to the control valve, wherein the housing includes a first chamber that accommodates the power transmission mechanism and the first pump, and a second chamber that is arranged adjacent to the first chamber in a horizontal line direction, in the first chamber, the first pump is arranged such that an axis of rotation of the first pump extends along a direction of an axis of rotation of the power transmission mechanism, in the second chamber, the control valve is arranged upright, and in the second chamber, the second pump is arranged upright with an axis of rotation of the second pump extending along a vertical direction.
2. The power transmission device according to claim 1, wherein the control valve and the second pump are aligned in the direction of the axis of rotation of the power transmission mechanism in the second chamber.
3. The power transmission device according to claim 2, wherein the second chamber is located toward a front of the vehicle as viewed from the first chamber, as viewed from the front of the vehicle, the control valve has a cut-out, and as viewed from the front of the vehicle, the second pump is arranged at least partially in the cut-out.
4. The power transmission device according to claim 3, wherein as viewed from the front of the vehicle, the cut-out is provided at a bottom of the control valve, and as viewed from one side in the

vertical direction, the second pump and the control valve are provided in an overlapping positional relationship.

5. The power transmission device according to claim 4, wherein the second pump includes a motor unit housing a motor, and a pump unit configured to suction and pressurize the oil using a rotational force of the motor, the motor unit and the pump unit are aligned in a direction of the axis of rotation of the second pump, as viewed from the front of the vehicle, the second pump is provided so that an axis of rotation of the motor extends along a direction orthogonal to the axis of rotation of the power transmission mechanism.

6. The power transmission device according to claim 5, wherein the pump is located below the motor unit.

7. The power transmission device according to claim 1, further comprising a strainer provided in the first chamber, wherein the first pump and the second pump share the strainer.

8. The power transmission device according to claim 7, wherein as viewed from the front of the vehicle, the control valve is provided in a positional relationship overlapping the first pump and the strainer.
